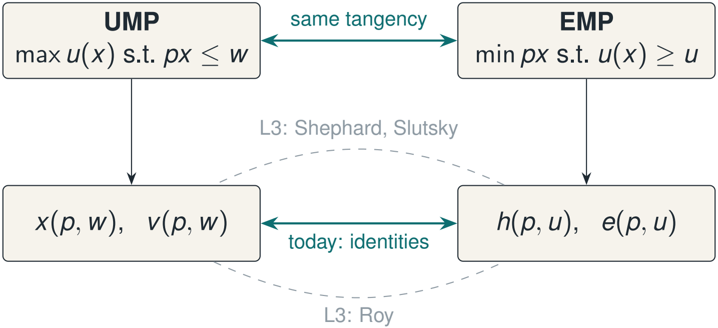
Lecture 2 · Classical Demand I

MWG Ch. 3.A–3.E · two problems, four objects, one square

Advanced Microeconomics 1

Thursday 24 September

The map for the next two lectures



- Today: build all four corners and connect them by *level* identities. L3: connect by *derivatives*. L4: run the arrows backwards (integrability) and cash out (welfare).
- **Why grind through this?** Both goals spend today's structure: goal (a)'s demand-function test lives in the derivatives of these objects (L3), and goal (b)'s welfare measures *are* differences of $e(p, u)$ (L4). Nothing on today's board is decoration.

Preference axioms for markets

Desirability

Throughout Ch. 3: $X = \mathbb{R}^L_+$, $\succsim$ rational. Add structure:

Definition · desirability axioms

- **Monotone**: $y \gg x \Rightarrow y \succ x$. **Strongly monotone**: $y \geq x,\ y \neq x \Rightarrow y \succ x$.
- **Locally nonsatiated (LNS)**: for every x and $\varepsilon > 0$ there is y with $\|y - x\| < \varepsilon$ and $y \succ x$.

- LNS is the weakest and the workhorse: it allows bads and thick regions of indifference in some directions, but rules out *thick indifference sets* — and it is exactly what delivers Walras' law.
- Hierarchy: strongly monotone $\Rightarrow$ monotone $\Rightarrow$ LNS.

Convexity

Definition · convexity

$\succsim$ is **convex** if the upper contour set $\{y : y \succsim x\}$ is convex for every x; **strictly convex** if $y \succsim x$, $z \succsim x$, $y \neq z$ imply $\alpha y + (1 - \alpha)z \succ x$ for $\alpha \in (0, 1)$.

- Interpretations: diminishing marginal rates of substitution; a taste for diversification (averages beat extremes among acceptable bundles).
- What it buys downstream: convex-valued demand; strict convexity $\Rightarrow$ single-valued, hence *functions* rather than correspondences.
- Honest note: convexity is an assumption of convenience. Nothing in “rationality” requires it; nonconvex tastes (specialization, addiction thresholds) simply produce jumpy demand.

Two families you will use constantly

Homothetic

Indifference sets are radial blow-ups: $x \sim y \Rightarrow \alpha x \sim \alpha y$. Representable by a u.h.d.1 utility.

Buys: linear wealth expansion paths, $x(p, w) = w \cdot x(p, 1)$ — the aggregation-friendly case (Gorman, EC2).

Quasilinear (in good 1)

$x \sim y \Rightarrow x + \alpha e_1 \sim y + \alpha e_1$; representable as $u(x) = x_1 + \phi(x_2, \ldots, x_L)$.

Buys: zero wealth effects on goods $2, \ldots, L$; CV $=$ EV $=$ consumer surplus — the welfare-friendly case (L5).

- Pattern: special preference classes are not curiosities; each one *switches off* a complication (wealth effects, distribution dependence) so a particular application becomes exact.

Continuity and Debreu's theorem

Definition · continuity

$\succsim$ is continuous if it is preserved under limits: for any sequences $x^n \to x$, $y^n \to y$ with $x^n \succsim y^n$ for all n, we have $x \succsim y$. (Equivalently: upper and lower contour sets are closed.)

Proposition · existence (MWG 3.C.1)

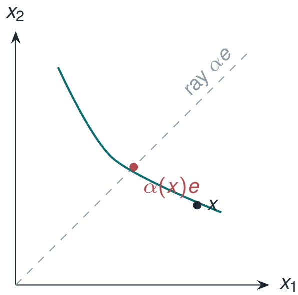
If $\succsim$ is rational and continuous on $X = \mathbb{R}^L_+$, it is represented by a *continuous* utility function.

- The counterexample continuity excludes: **lexicographic** preferences. Rational, strongly monotone — but each x_1-value would need its own non-degenerate utility interval, and $\mathbb{R}$ has only countably many disjoint ones. No representation exists.

The one-slide construction (monotone case)

Assume $\succsim$ also monotone; let $e = (1, \ldots, 1)$.

1. For each x, the sets $A^+ = \{\alpha \geq 0 : \alpha e \succsim x\}$ and $A^- = \{\alpha \geq 0 : x \succsim \alpha e\}$ are closed (continuity), nonempty, and cover $\mathbb{R}_+$ (completeness).
2. $\mathbb{R}_+$ connected $\Rightarrow$ they intersect: some $\alpha(x)$ has $\alpha(x)e \sim x$. Monotonicity makes it unique.
3. Set $u(x) = \alpha(x)$: then $x \succsim y \iff \alpha(x) \geq \alpha(y)$. $\square$



- Measure each bundle by *how far up the diagonal its indifference curve crosses.* Elegant, constructive, and the template for money-metric utility later.

Dictionary: preference properties $\leftrightarrow$ utility properties sharpening hypothesis (a)

$\succsim$ ...	any representing u is ...
monotone	nondecreasing (increasing if strongly monotone)
convex	**quasiconcave**
strictly convex	strictly quasiconcave
homothetic	has an h.d.1 representation
quasilinear in good 1	has a $u = x_1 + \phi(x_{-1})$ representation

- Quasiconcavity, not concavity: convex preferences pin down only the shape of level sets, and quasiconcavity is the ordinal (transformation-proof) notion. Concavity of u is a cardinal statement — meaningless until Ch. 6.

The Utility Maximization Problem

UMP: statement and existence

UMP

$$\max_{x \in \mathbb{R}^L_+} u(x) \quad \text{s.t.} \quad p \cdot x \leq w, \qquad p \gg 0.$$

Solution set: **Walrasian demand** $x(p, w)$; value: **indirect utility** $v(p, w)$.

- Existence: $B_{p,w}$ is compact ($p \gg 0$ does the work), u continuous $\Rightarrow$ Weierstrass. Note existence needs *no* convexity.
- This is the same object as Ch. 2's $x(p, w)$ — but now derived, so its properties come with reasons attached.

First-order conditions

With u differentiable, Kuhn–Tucker at an optimum x^*: there is $\lambda \geq 0$ with

$$\frac{\partial u}{\partial x_\ell}(x^*) \leq \lambda\, p_\ell, \quad \text{with equality if } x_\ell^* > 0.$$

- Interior solution: for any two goods consumed, $\mathrm{MRS}_{\ell k} = \dfrac{\partial u/\partial x_\ell}{\partial u/\partial x_k} = \dfrac{p_\ell}{p_k}$ — the tangency picture.
- Corner: MRS $\leq$ price ratio can hold with strict inequality for unconsumed goods. *Always check corners in problem sets; graders always put one in.*
- $\lambda =$ marginal utility of wealth: $\lambda = \partial v/\partial w$ (envelope — proved properly next lecture).
- FOCs are sufficient as well as necessary when u is quasiconcave with $\nabla u \neq 0$ (LNS).

Properties of Walrasian demand

Proposition · MWG 3.D.2

If u is continuous and represents LNS $\succsim$, then $x(p, w)$ satisfies:

1. **Homogeneity of degree 0** in (p, w) [budget set unchanged]
2. **Walras' law**: $p \cdot x = w$ for all $x \in x(p, w)$ [LNS: money left over $\Rightarrow$ better bundle nearby]
3. If $\succsim$ convex: $x(p, w)$ convex-valued; if strictly convex: **single-valued**.

- Note the bookkeeping: each Ch. 2 *assumption* is now a *theorem*, with a named axiom responsible. This is what the preference route buys: traceability.

Worked example: Cobb–Douglas (memorize the whole row)

$u(x) = x_1^{\alpha} x_2^{1-\alpha}$, $\alpha \in (0,1)$:

$$x_1(p,w) = \frac{\alpha w}{p_1}, \qquad x_2(p,w) = \frac{(1-\alpha)w}{p_2} \qquad \text{(constant expenditure shares)}$$

$$v(p,w) = w\,\alpha^{\alpha}(1-\alpha)^{1-\alpha}\,p_1^{-\alpha}p_2^{-(1-\alpha)}$$

- Sanity checks to run every time: h.d.0 in (p,w)? Walras' law? v increasing in w, decreasing in p?
- Cobb–Douglas is the special case where income and substitution effects on cross-demands exactly cancel ($\partial x_1/\partial p_2 = 0$) — convenient, and dangerously unrepresentative. CES (later today) is the honest laboratory.

Indirect utility

Proposition · MWG 3.D.3

$v(p, w)$ is: (i) h.d.0; (ii) strictly increasing in w, nonincreasing in each p_ℓ; (iii) continuous; (iv) **quasiconvex**: $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex.

- Quasiconvexity is the surprising one. Intuition: fix the optimal bundle $\bar{x}$ at (p, w). Averaging two price–wealth situations that each leave the consumer no better than $\bar{v}$ cannot open up new opportunities — the averaged budget set is contained in the union's "convexification" of constraints, not of choices.
- Practical use of v: it is the object welfare analysis differentiates (Roy, L4), and the object applied GE models parameterize.

The Expenditure Minimization Problem

EMP: the dual question

EMP

$$\min_{x \in \mathbb{R}^L_+} p \cdot x \quad \text{s.t.} \quad u(x) \geq u \quad (u > u(0),\ p \gg 0).$$

Solution set: **Hicksian (compensated) demand** $h(p, u)$; value: **expenditure function** $e(p, u)$.

- Why bother with a problem no consumer solves? Because *welfare questions are expenditure questions*: "how many dollars compensate this price change?" is literally $e(\cdot)$ evaluated twice (L5).
- And because e has better mathematical manners than v (concavity in p) — which is where all the derivative magic comes from.

Properties of Hicksian demand

Proposition · MWG 3.E.3

For continuous u representing LNS $\succsim$: $h(p, u)$ is (i) h.d.0 *in* p; (ii) exhausts the constraint: $u(x) = u$ for $x \in h(p, u)$ ("no excess utility"); (iii) convex-/single-valued under convexity/strict convexity.

Proposition · compensated law of demand — now for free

For any p', p'': $(p'' - p') \cdot [h(p'', u) - h(p', u)] \leq 0$.

Two-line proof. $h(p'', u)$ minimizes cost at p'' among bundles reaching u, and $h(p', u)$ is such a bundle: $p'' h(p'', u) \leq p'' h(p', u)$. Symmetrically at p'. Add. $\square$

- Own-price Hicksian demand never slopes up: Giffen behavior is an income effect wearing a costume. No compensation subtlety needed here — unlike Ch. 2.

The expendure function

toolkit for (b) and recovery

Proposition · MWG 3.E.2

$e(p, u)$ is: (i) h.d.1 in p; (ii) strictly increasing in u, nondecreasing in each p_ℓ; (iii) continuous; (iv) **concave in** p.

Concavity, the proof to internalize. Fix u; take $p^\alpha = \alpha p' + (1-\alpha)p''$ and let $x^\alpha \in h(p^\alpha, u)$. Then

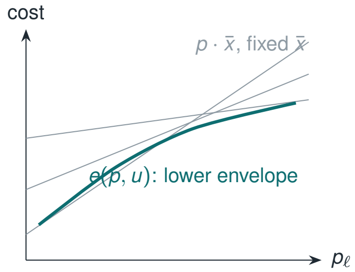
$$e(p^\alpha, u) = \alpha\, p' x^\alpha + (1-\alpha)\, p'' x^\alpha \;\geq\; \alpha\, e(p', u) + (1-\alpha)\, e(p'', u),$$

since x^α is feasible (reaches u) at both p' and p'' but not necessarily optimal. □

- **Why e is the block's most valuable object:** in 1981 Hausman computed the *exact* welfare cost of a tax from one estimated demand curve — this block's paper for goal (b). Every input he needed is now on this board; the theorem that assembles them arrives Monday.

Why concavity: the envelope-of-lines picture

- For each *fixed* bundle x with $u(x) \geq u$, the map $p \mapsto p \cdot x$ is **linear**.
- $e(\cdot, u)$ is the **pointwise minimum** over all these linear functions — a lower envelope of hyperplanes. Lower envelopes of linear functions are concave. Always.
- Economic content: a *passive* consumer (fixed bundle) has linear costs; *re-optimization* can only help as prices move away — bending the true cost *below* the tangent line.



One trick, three functions

This exact argument will reappear for the *profit function* $\pi(p)$ (convex: upper envelope,

Connecting the corners: the duality identities

Proposition · MWG 3.E.1 and companions

Under LNS and continuity, for $p \gg 0$, $w > 0$, attainable u:

$$e\big(p,\ v(p,w)\big) = w, \qquad v\big(p,\ e(p,u)\big) = u,$$
$$h(p,u) = x\big(p,\ e(p,u)\big), \qquad x(p,w) = h\big(p,\ v(p,w)\big).$$

- Same tangency, two descriptions: UMP fixes the budget and finds the highest indifference curve; EMP fixes the indifference curve and finds the lowest budget. LNS guarantees neither problem leaves slack.
- These four identities are the workhorse of every derivation in L4: Slutsky is literally "differentiate the third one."

The CES laboratory (two goods)

$u(x) = \left(x_1^\rho + x_2^\rho\right)^{1/\rho}$, $\rho < 1$, $\rho \neq 0$; elasticity of substitution $\sigma = 1/(1-\rho)$. Define the price index $P \equiv \left(p_1^{1-\sigma} + p_2^{1-\sigma}\right)^{1/(1-\sigma)}$.

Walrasian demand	$x_\ell(p, w) = \dfrac{p_\ell^{-\sigma}}{P^{1-\sigma}}\, w$
Indirect utility	$v(p, w) = w/P$
Expenditure	$e(p, u) = u \cdot P$
Hicksian demand	$h_\ell(p, u) = u \cdot p_\ell^{-\sigma} P^\sigma$

- Limits: $\sigma \to 1$ Cobb–Douglas; $\sigma \to 0$ Leontief; $\sigma \to \infty$ linear. PS1 has you verify every duality identity on this family — do it once by hand, and L4 will feel inevitable.

Scorecard, takeaways, readings

Takeaways

- **Goal (a):** the hypothesis is now operational — continuity (not rationality) buys representation; convexity buys well-behaved demand; LNS buys Walras' law. What each assumption purchases, itemized.
- **Goal (b):** its objects now exist — x, v, h, e linked by level identities, e concave by the envelope of lines — but cannot yet be reached *from data*. Monday's theorem closes that gap.
- Hicksian demand obeys the law of demand unconditionally; Giffen is an income effect — the descriptive layer, upgraded.

Required: MWG 3.A–3.E. **Next:** 3.F–3.G before Thursday; start Hausman (1981), this block's (b) paper.

The practical is a week today: all it needs is a laptop with a working R installation — check yours starts now, not the morning of.